



PK FOKAM INSTITUTE OF EXCELLENCE

Dignity – Faith – Responsibility

DEPARTMENT OF COMPUTER & MECHANICAL ENGINEERING

**Project: Development of a Restaurant Management Web
Application**

REPORT v1.2

COURSE: INTRODUCTION TO PYTHON

THEME: PREPARATION AND RESEARCH FOR THE DEVELOPMENT OF A RESTAURANT WEB
APP

Group Members:

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Work Distribution

The project is carried out by a team of four members. Each member has a specific role in the project while contributing to the tasks required at each

Table 1: General role

Member	Role	Matricule number
Kamdem Matchum Judith Purchase	Database Analysis, Documentation, Project Coordination	25S052
Mvondo Mvondo Wilfried Anthony	Backend Developer	25S048
Massado Djoko Aicha	Frontend Developer	26S003
Mep Audrey Larissa	System Integration and Testing	26S014

Contribution During Stage 1 and 2 (Current Report)

Since this report covers *the first stage of the project*, the following tasks were completed collaboratively by the team:

Member	Contribution	Percentage
Kamdem Matchum Judith Purchase	Analyzed the SQL database structure, studied and understood most of the system models, and prepared the project documentation and report writing	25%
Mvondo Mvondo Wilfried Anthony	Assisted in setting up the Django development environment and project structure, and developed the Class Diagram and Gantt Chart	25%
Massado Djoko Aicha	Participated in studying the system requirements, planned the application interface, and created the Use Case Diagram, Sequence Diagram, and Gantt Chart	25%
Mep Meku Audrey	Helped configure development tools, assisted in project planning, and participated in the analysis of the Class Diagram	25%

Table 1.1: Role for stage 1&2

1. Introduction

The purpose of this project is to design and develop a restaurant management web application that will help restaurants manage their daily operations efficiently. The system will allow restaurant staff to manage customers, orders, reservations, employees, and inventory through a centralized web interface.

The application will be developed using Python and the Django framework, providing a secure and scalable backend environment. A relational database will store restaurant information such as customer data, menu items, orders, and stock.

This project is carried out as a group assignment and will be completed in three stages, each accompanied by a progress report.

2. Project Objectives

Main Objective: Develop a functional web-based system that simplifies restaurant management tasks.

Specific objectives:

- Design a database system to manage restaurant data
- Develop backend logic using Django

- Create a user-friendly web interface for staff
- Manage restaurant operations: customer info, orders, reservations, employees, inventory
- Ensure the system is organized, efficient, and easy to use

3. Project Phases

The development of the Restaurant Management Web Application is divided into three main phases.

- Phase 1: Project Preparation and Initial Setup

This phase focuses on understanding the project requirements and preparing the development environment. It includes installing the required tools, creating the Django project structure, analyzing the SQL database, and connecting the application to the database.

- Phase 2: System Development

This phase focuses on building the main functionalities of the application. It includes developing the backend logic using Django, designing the user interface, and implementing the core modules such as customer management, orders, reservations, employees, and inventory.

- Phase 3: System Integration and Testing

This phase focuses on integrating all system components and ensuring that the application works correctly. It includes testing the system, fixing errors, improving performance, and preparing the final version of the application.

4. System Analysis and Description of the Proposed System

The proposed system is a restaurant management web application enabling administrators and staff to manage restaurant operations.

Modules:

- Customer Management: Store and track customer information
- Order Management: Record and manage customer orders
- Reservation Management: Handle table reservations
- Employee Management: Store employee information
- Inventory and Stock Management: Monitor ingredients and stock levels

All modules interact with a central database for organized and accessible information.

5. Tools and Technologies

- Python: Backend programming language



Figure a: Python app

- Django Framework: Handles database, web pages, and security



Figure b: Django app

- MySQL Database: Stores restaurant data



Figure c: Database tool

- GitHub: Version control and collaboration

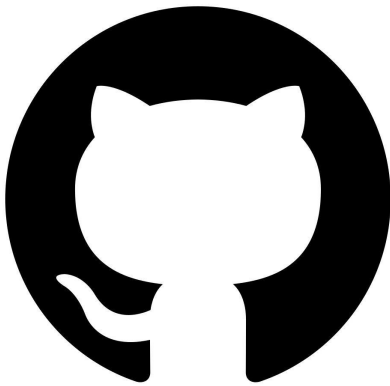


Figure d: Github

- Web Technologies: HTML, CSS, possibly JavaScript for frontend



Figure e: Html app

6. Database Overview

The database stores all information required for restaurant operations.

Tables:

- Customers
- Orders
- Reservations
- Employees
- Products/Menu items
- Stock/Inventory

Sample

C:\Windows\system32\cmd.exe - mysql -u root -p

```

+-----+
| 1840000 |
+-----+
1 row in set (0.00 sec)

+-----+
| Nom | Prenom | Tel | Poste |
+-----+
| Fokou | Alain | 699887766 | Directeur General |
| Mballa | Serge | 677445566 | Chef Cuisinier |
| Essomba | Jean | 696554433 | Chef Cuisinier |
| Nguetack | Linda | 655778899 | Serveur |
| Tchoua | Patrick | 670998877 | Agent de Marketing |
| Pokam | Justin | 677809865 | Chauffeur |
| Kenfack | Paul | 672801865 | Technicien |
| Fotso | Mirielle | 655879034 | Cuisinier |
| Ngo | Alice | 655833031 | Cuisinier |
| Nyjng | Ella | 655119030 | Cuisinier |
| Kamani | Diane | 685820131 | Cuisinier |
| Tchuene | Eric | 675879047 | Chef du Magasin |
| Ousmane | Bakary | 675449044 | Comptable |
| Talla | Jean-Luc | 675867041 | Gestionnaire de Stock |
| Nyjmen | Anderson | 675060048 | Responsable des Achats |
| Asha | Maeva | 681603279 | Agent T#1 #phonique |
| Zogo | Brice | 690112233 | Agent de S#curit# |
| Zanga | Merlin | 690551403 | Chauffeur |
+-----+

18 rows in set (0.00 sec)

+-----+
| id_Commande | Ingredient | Prix_Commande |
+-----+
| 1 | Cube d'assaisonnement | 6500 |
| 3 | Cube d'assaisonnement | 12000 |
| 1 | Gingembre | 6500 |
| 3 | Gingembre | 12000 |
| 1 | Huile de Soja | 6500 |
| 2 | Huile de Soja | 8000 |
| 3 | Huile de Soja | 12000 |
| 4 | Huile de Soja | 9000 |
| 6 | Huile de Soja | 4500 |
+-----+

```

Figure f: Data in the sql file

The design ensures organized, consistent, and retrievable data. The SQL file is included with this report.

7. Project Planning:

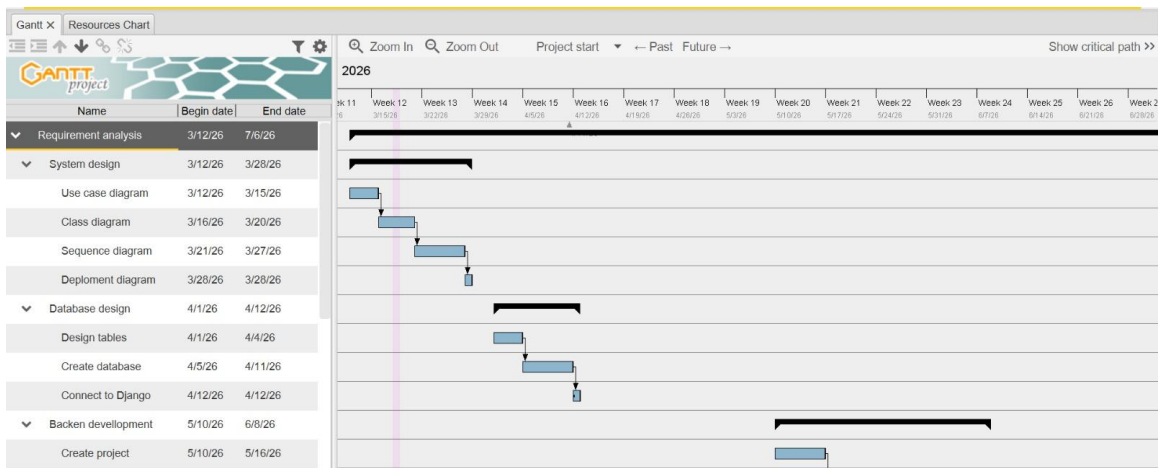


Figure 7: Gantt Chart

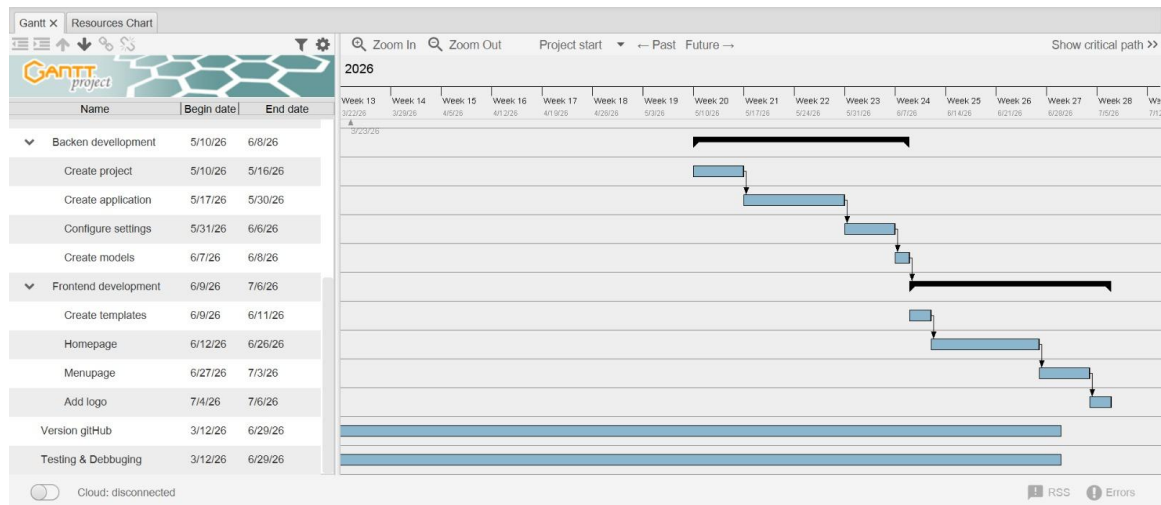


Figure 8: Gantt chart

8. System Modeling

To better understand the structure and behavior of the system, several UML diagrams were

developed.

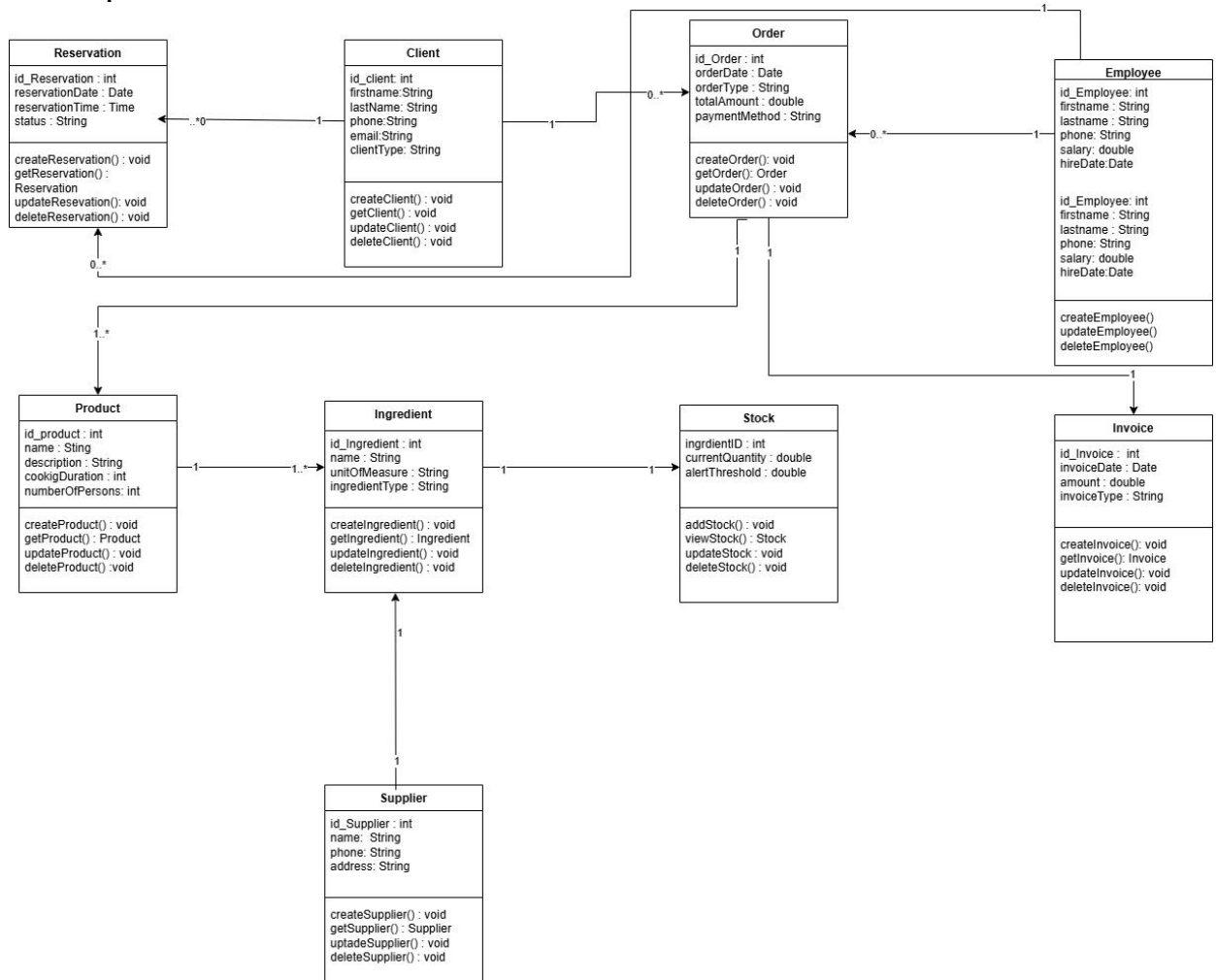


Figure 9: Class model

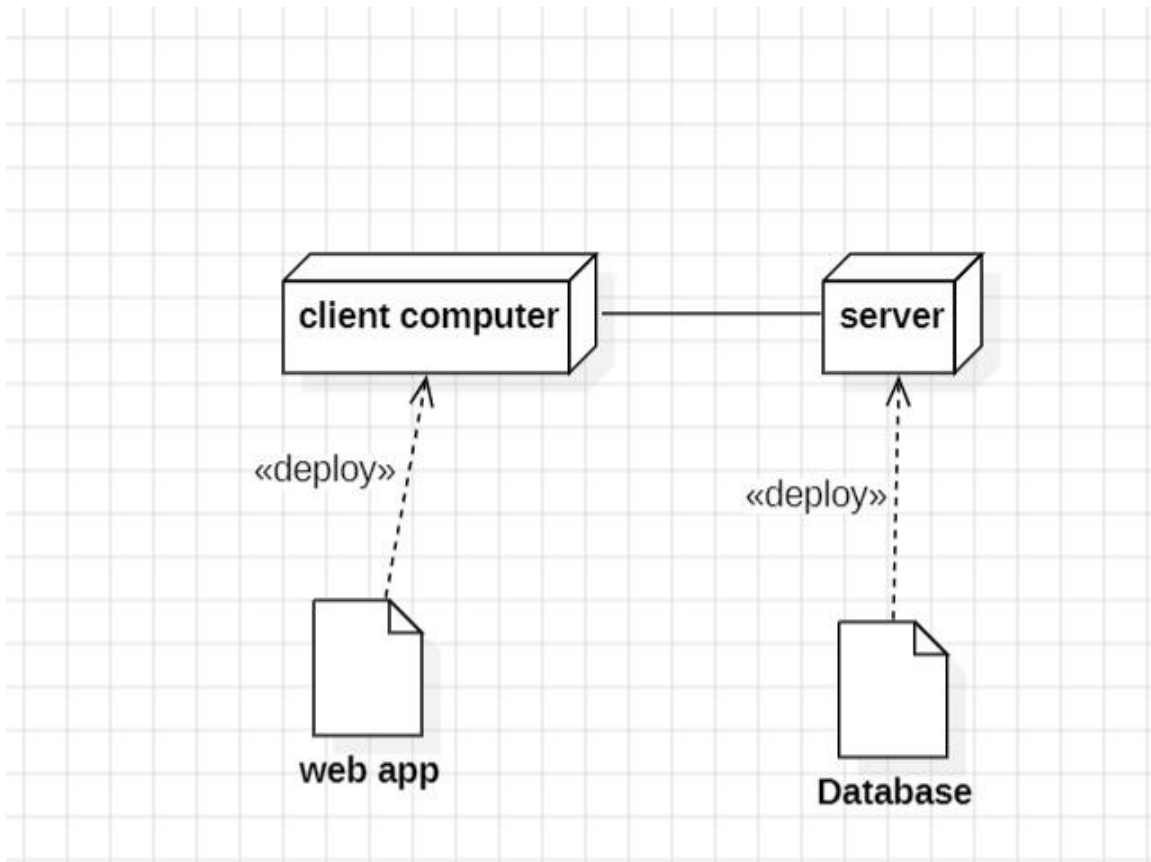


Figure 10: Deployment Diagram

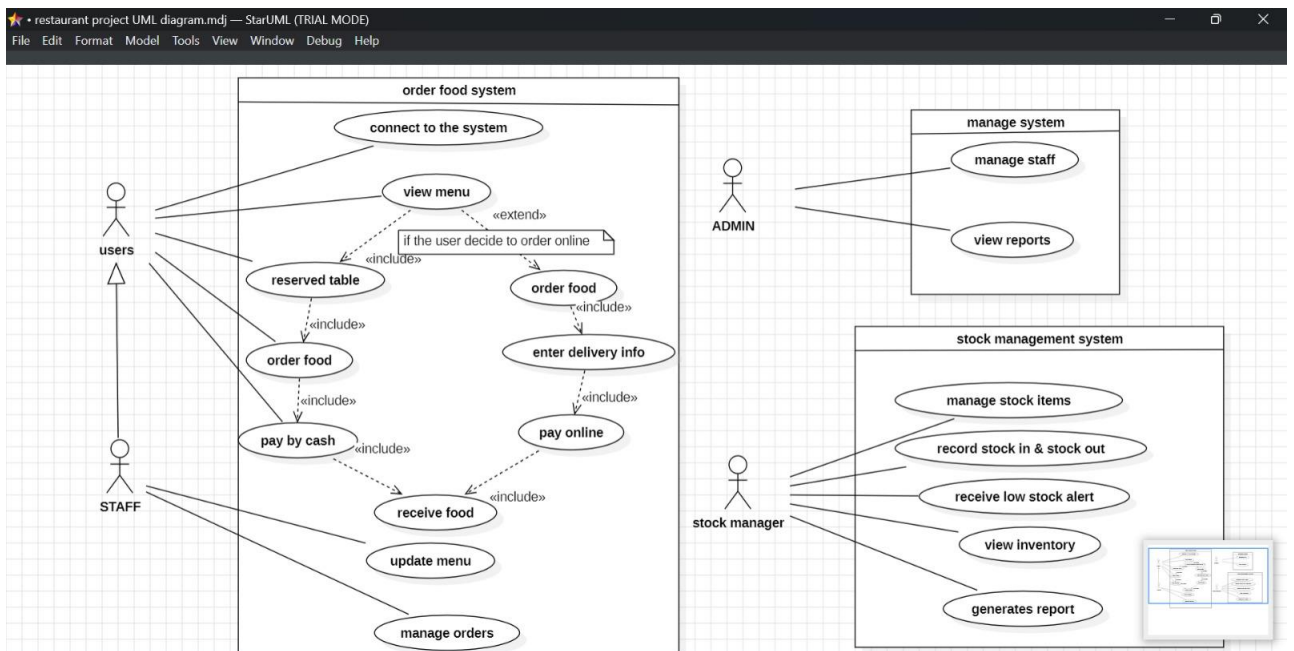


Figure k: User case diagram

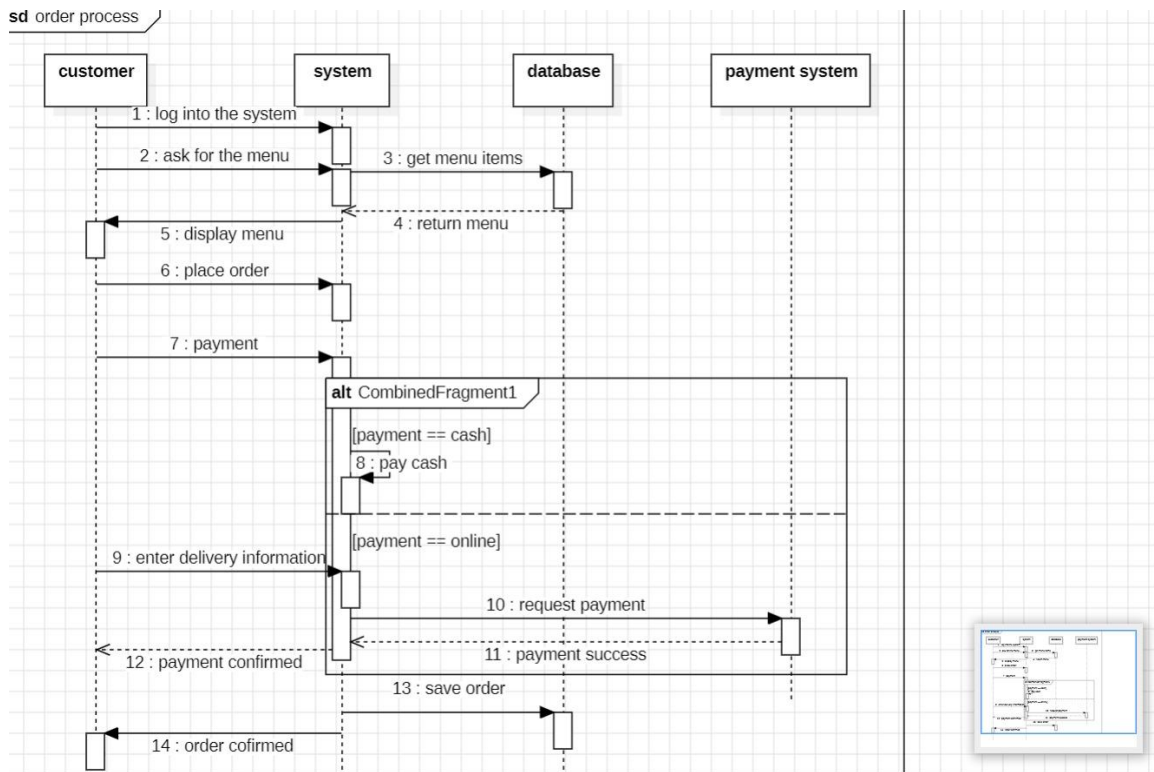


Figure 12: Sequence Diagram

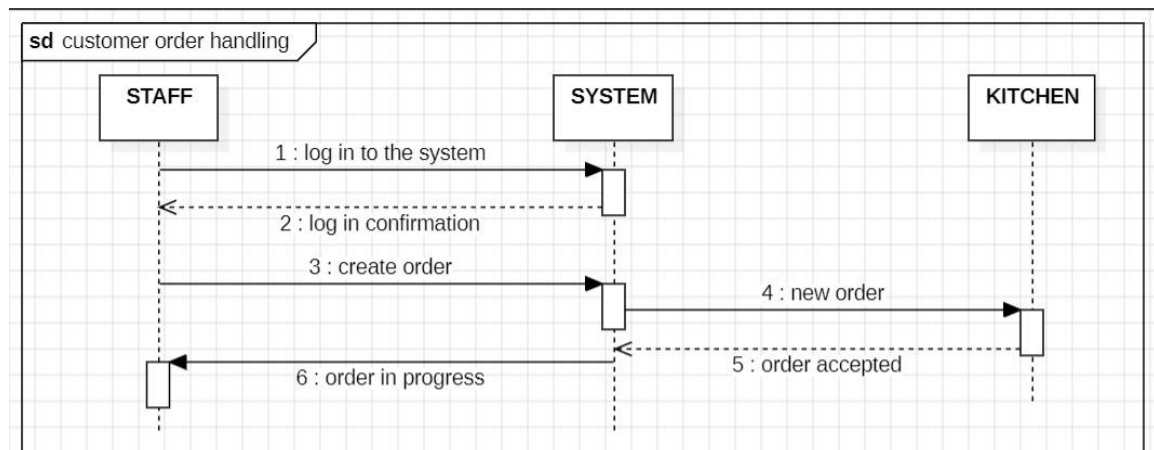


Figure 13: Sequence diagram continuity

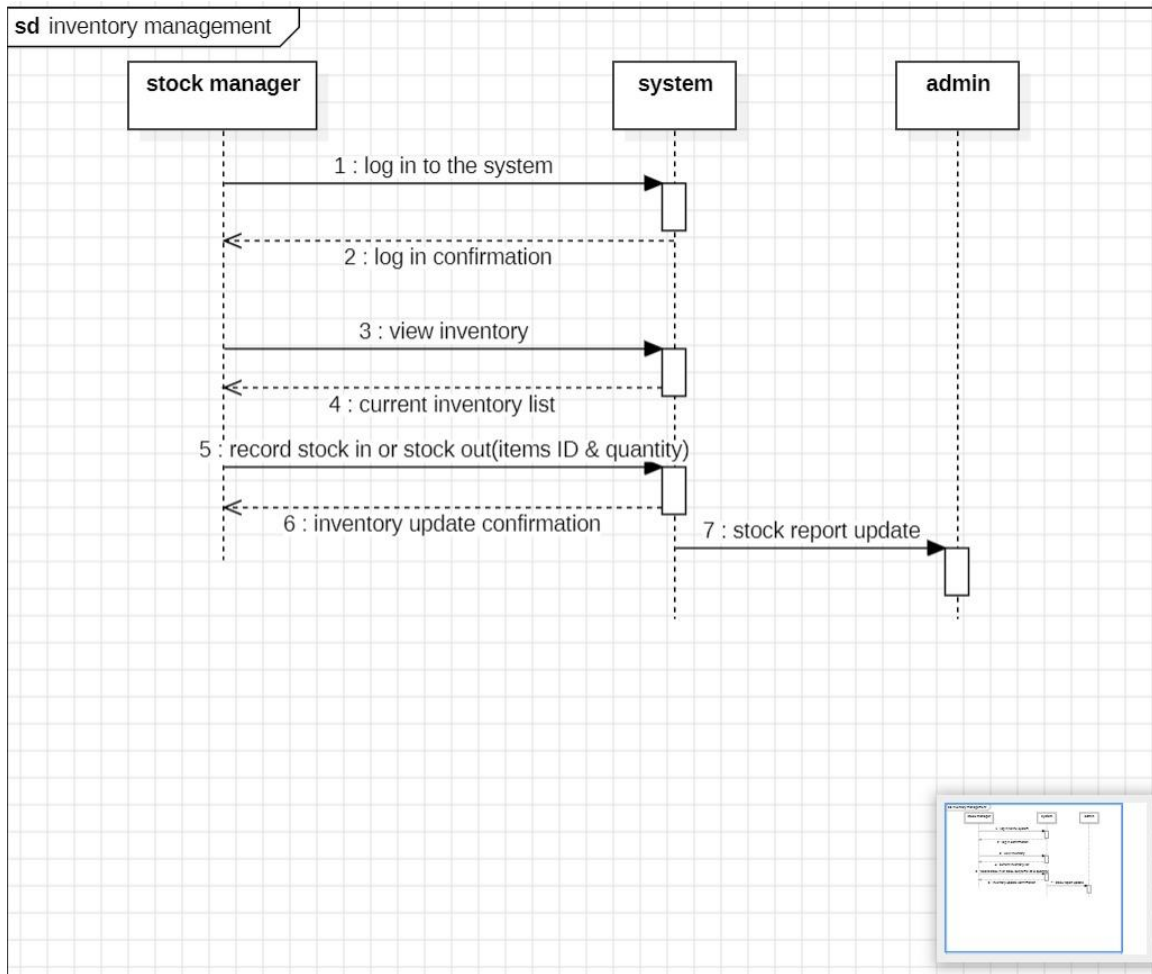


Figure 14: Sequence diagram end

9. Current Progress of the Project

Completed tasks:

- Installed and configured development tools (Python, Django)

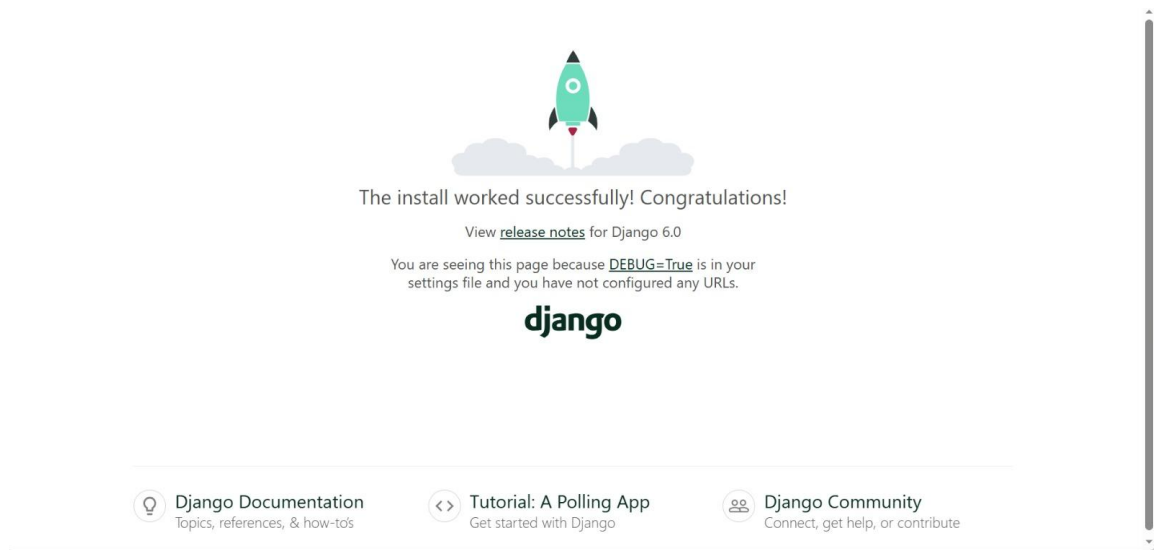


Figure 15: Django complete installation

- Created GitHub repository

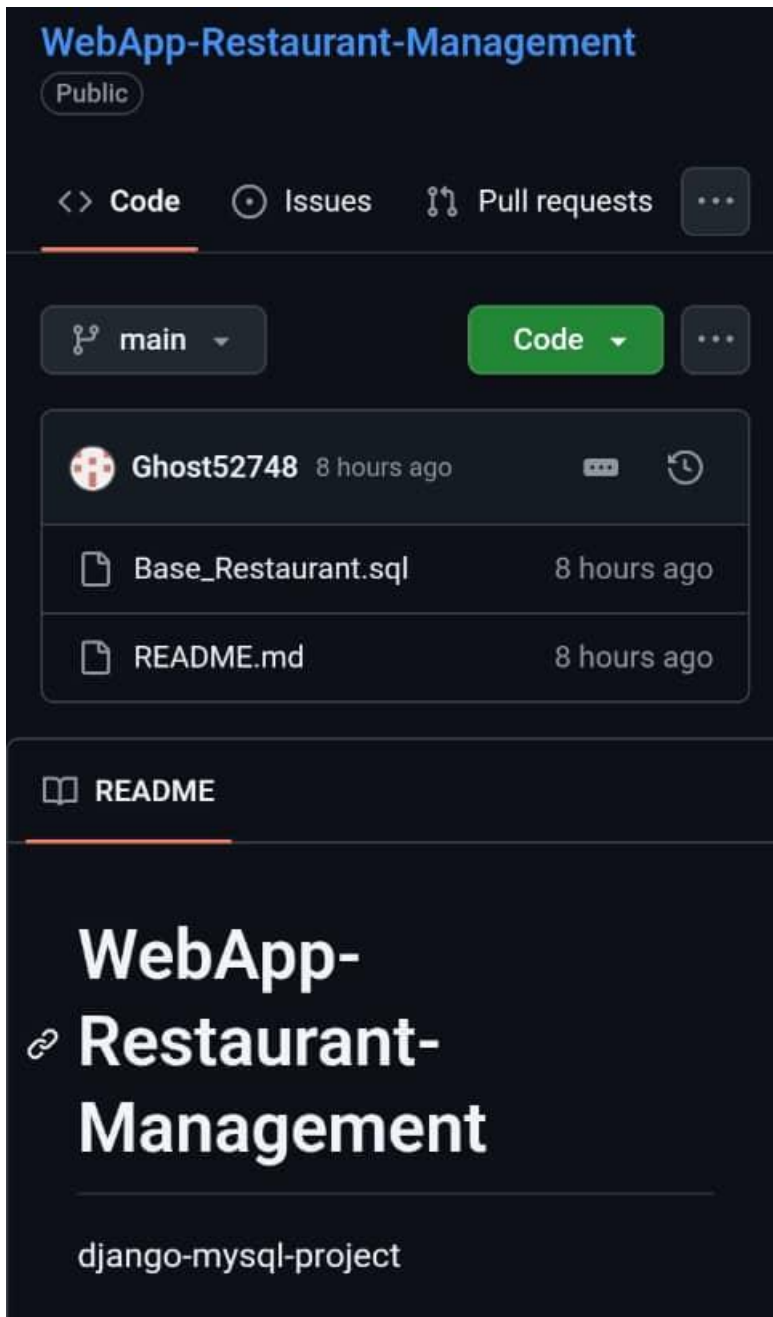


Figure 16: Github repository

- Created Django project structure



Figure 17: Project structure

- Implemented database models in Django

```
C:\Windows\system32\cmd.exe
(venv) C:\Users\admin\django_project\config>notepad settings.py

(venv) C:\Users\admin\django_project\config>cd ..

(venv) C:\Users\admin\django_project\config>python manage.py dbshell
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 51
Server version: 8.0.40 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW TABLES
+-----+
| Tables_in_restaurant_db |
+-----+
| auth_group               |
| auth_group_permissions   |
| auth_permission          |
| auth_user                |
| auth_user_groups         |
| auth_user_user_permissions |
| django_admin_log         |
| django_content_type      |
| django_migrations        |
| django_session           |
+-----+
10 rows in set (0.00 sec)

mysql> exit;
Bye

(venv) C:\Users\admin\django_project\config>python manage.py migrate admin
Operations to perform:
  Apply all migrations: admin
```

Figure 18: Database models in Django

- Connected Django application to the database

Database link with Django

```

1340000
1 row in set (0.00 sec)

```

Nom	Prenom	Tel	Poste
Fokou	Alain	699887766	Directeur General
Mballa	Serge	677445566	Chef Cuisinier
Essomba	Jean	696554433	Chef Cuisinier
Nguefack	Linda	655778899	Serveur
Tchoua	Patrick	670998877	Agent de Marketing
Pokam	Justin	677809865	Chauffeur
Kenfack	Paul	672801865	Technicien
Fotso	Miriel	655879034	Cuisinier
Ngo	Alice	655833821	Cuisinier
Njing	Ella	655119030	Cuisinier
Kameni	Diane	685829131	Cuisinier
Tchuene	Eric	675879047	Chef du Magasin
Ousmane	Bakary	675449044	Comptable
Talla	Jean-Luc	675867041	Gestionnaire de Stock
Nyimen	Anderson	675060048	Responsable des Achats
Asha	Maeve	681603279	Agent T ^{#1} #phonique
Zogo	Brice	690112233	Agent de S [#] curit [#]
Zanga	Merlin	690551403	Chauffeur

```

18 rows in set (0.00 sec)

```

id_Commande	Ingredient	Prix_Commande
1	Cube d'assaisonnement	6500
3	Cube d'assaisonnement	12000
1	Gingembre	6500
3	Gingembre	12000
1	Huile de Soja	6500
2	Huile de Soja	8000
3	Huile de Soja	12000
4	Huile de Soja	9000
6	Huile de Soja	4500

Figure 19: Database link with Django

- Project plan using gantt chart
- Understanding and representing the management system using models

In progress:

- Study Django framework more deeply
- Analyze SQL database structure and table relationships
- Continuous improvement of system architecture and project plan

10. Difficulties Encountered

During the first phase of the project, the group encountered several technical and organizational challenges:

- Learning the Django framework for some members who were not previously familiar with it

- Understanding the database structure due to the presence of multiple related tables
- Difficulty understanding and creating UML diagrams such as Class, Use Case, and Sequence diagrams
- Installing and configuring development tools such as Python, Django, StarUML, and GanttProject
- Coordinating tasks and planning work effectively among group members
- Understanding how to properly analyze system requirements before designing the models

11. Solutions Applied

To overcome these challenges, the group adopted the following solutions:

- Members studied Django documentation, watched tutorials, and practiced small examples to improve their understanding of the framework
- The group analyzed the SQL database structure together to better understand relationships between tables and how data flows in the system
- Members researched UML modeling concepts and collaborated to better understand how to design Class, Use Case, and Sequence diagrams
- Members who successfully installed the required tools assisted others in configuring their development environment
- Tasks were clearly divided among members according to roles and competencies to improve efficiency
- Regular communication and group discussions were used to clarify requirements and improve project planning

12. Next Steps and Implementation Plan

- Develop backend logic
- Design web interface
- Implement core functionalities: orders, customers, reservations
- Testing and integration of modules

13. Conclusion

Stage 1 focused on environment preparation, initial implementation, and database analysis. Challenges were addressed, workflow organized, and foundation laid for full development.

14. GitHub Repository Link

<https://github.com/Ghost52748/WebApp-Restaurant-Management>

15. SQL Database File

The SQL file containing the database structure is included in our Github repository as part of the project submission.

```

DROP DATABASE IF EXISTS Restaurant;
CREATE DATABASE Restaurant CHARACTER SET utf8mb4 COLLATE utf8mb4_general_ci;
USE Restaurant;
/* ===== TABLES ===== */
CREATE TABLE Client (
  id_Client INT PRIMARY KEY,
  Nom VARCHAR(20),
  Prenom VARCHAR(20),
  Tel VARCHAR(15),
  Email VARCHAR(45),
  Type_de_Client VARCHAR(20)
);
CREATE TABLE `Table` (
  id_Table INT PRIMARY KEY,
  Numero_Table VARCHAR(3),
  Capacite INT,
  Commentaire TEXT
);
CREATE TABLE Reservation (
  id_Reservation INT PRIMARY KEY,
  id_Client INT,
  id_Table INT,
  `Date` DATE,
  Heure TIME,
  Statut VARCHAR(45),
  FOREIGN KEY (id_Client) REFERENCES Client(id_Client),
  FOREIGN KEY (id_Table) REFERENCES `Table`(id_Table)
);
CREATE TABLE Ingredient (
  id_Ingredient INT PRIMARY KEY,
  Nom VARCHAR(45),
  Unite_de_Mesure VARCHAR(45),
  `Type` VARCHAR(30)
);

```

Figure 20: Database file